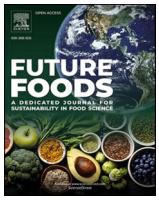
Future Foods 14 (2026) 101237



Contents lists available at ScienceDirect

Future Foods

journal homepage: www.elsevier.com/locate/fufo



Impact of nixtamalization processing on the rheological, nutritional, and physicochemical properties of tortillas made from nixtamalized maize–bean flours

Rosa María Mariscal-Moreno [a,*], Juan de Dios Figueroa-Cárdenas [b], Cristina M. Rosell [c,d], Nicola Gasparre [c], José Juan Véles-Medina [b]

[a] Departamento de Salud, Universidad Iberoamericana, Ciudad de Mexico, Mexico. Prol. Paseo de la Reforma 880, A. Obregon Mexico C. P, 01219, Mexico
[b] Centro de Investigación y de Estudios Avanzados Unidad Querétaro, Libramiento Norponiente 2000, Fracc. Real de Juriquilla, Querétaro, Qro., C.P. 76230, Mexico
[c] Department of Food and Human Nutritional Sciences, University of Manitoba, Winnipeg, MB, R3T 2N2, Canada
[d] Institute of Agrochemistry and Food Technology (IATA-CSIC), Paterna, Spain

ARTICLE INFO

Keywords:
Nixtamalization
Bean fortification
Mineral content
Maize tortillas

ABSTRACT

This study evaluated the effects of traditional nixtamalization (TN), ecological nixtamalization (EN), and water cooking processing (CK) applied to four beans on the nutritional and physicochemical properties of composite tortillas produced from blends of bean and maize nixtamalized flours (50:50). Nixtamalization was the main driver of calcium enrichment, while bean type significantly influenced protein (14.61–18.11 g/100 g), crude fiber (2.78–3.39 g/100 g), carbohydrates (72.9–79.59 g/100 g), and ash contents. Traditional nixtamalization produced the lowest fiber content retention, whereas CK formulations showed the highest fiber and fat levels. Bean incorporation reduced total carbohydrates by up to 6% relative to maize tortillas (83.90 g/100 g) without major changes in energy density (394.8–398.7 kcal/100 g). Compared with flour tortillas, bean-added tortillas showed markedly higher iron and zinc contents, while ash increased slowly. Color attributes were changed by processing and formulation, reflecting pH-driven pigment changes. Nixtamalization process affected starch hydrolysis kinetics and resistant starch formation, whereas bean type affected matrix viscosity. Calcium chloride promoted rapid hydrolysis (k up to 1.40 min^{-1}) and the highest resistant starch levels (up to 5.04 g/100 g). Textural analysis showed that nixtamalized bean tortillas were softer than flour controls, indicating that bean type and nixtamalization treatment influenced tortilla textural properties.

1. Introduction

Maize (*Zea mays L.*) stands as a foundational crop globally. In Mexico and Central America, maize is the key ingredient for the traditional tortillas. The maize tortilla transcends its function as a mere food item, embodying deep social and cultural significance while driving significant economic activity. Statista (2026) reported that in 2024, the production of corn tortillas and nixtamalization milling in Mexico amounted to >90 billion Mexican pesos. The traditional method for preparing maize products is nixtamalization, an ancient alkaline thermal process that transforms raw kernels into dough and flour suitable for tortillas. The most common salt used for nixtamalization is calcium hydroxide ($Ca(OH)_2$, or lime). The calcium salt used in nixtamalization affects chemical transformations, including starch gelatinization, removal of the pericarp, and modifications of the protein solubility and structure (Rincón-Aguirre et al., 2021). Likewise, it significantly affects starch properties and digestibility; thus, it can influence the rate of glucose absorption (Mariscal Moreno et al., 2015). Furthermore, a relevant finding from nixtamalized products is the marked increase in calcium content (Escalante-Aburto et al., 2020). In addition, alternative processing approaches have been developed to reduce the environmental impact associated with alkaline wastewater (nejayote) generated during processing. Calcium salts such as calcium chloride or calcium carbonate operate at a lower alkalinity than lime and therefore modify the physicochemical transformations occurring during processing, including mineral uptake, starch structural changes and dietary fiber retention.

Although maize plays a pervasive role in the diet, its nutritional

* Corresponding author.
E-mail address: rosa.mariscal@ibero.mx (R.M. Mariscal-Moreno).

https://doi.org/10.1016/j.fufo.2026.101237
Received 7 June 2026; Received in revised form 9 October 2026; Accepted 27 October 2026
Available online 28 October 2026
2666-8335/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

profile is limited. Maize-based foods are typically low in total protein and dietary fiber relative to recommended intake levels, which may contribute to protein–energy malnutrition and micronutrient deficiencies in populations with high maize dependence (Sánchez-Villa et al., 2020). In this context, the incorporation of beans into maize-based products has been proposed as a strategy to improve their nutritional quality by increasing protein, fiber, and mineral content (Hassan et al., 2024; Cortes et al., 2023). Considering the high intake of tortillas and the associated nutritional deficiencies, it is relevant to emphasize their adequate contributions to daily nutrient intake. Accordingly, strategies aimed at increasing the macronutrient density of maize tortillas remain essential.

Pulses like beans are recognized globally as complex sources of carbohydrates (55–60%), protein (17–40%), fat, and high levels of dietary fiber (15–39%) (Alcázar-Valle et al., 2021). The incorporation of bean flour, such as common beans, chickpea, or soybean, into maize-based products have been proposed as an affordable and sustainable strategy to enhance macro- and micronutrient content (Boukid et al., 2019). This practice is supported by the global trend toward increasing the consumption of plant proteins to protect the environment and reduce carbon footprints (Boukid et al., 2019). Blending maize flour with pulses flour substantially enhances the macronutrient quality of tortillas, resulting in composite tortillas with protein contents far higher than those made solely from maize (Hernández-Salazar et al., 2006). Similar trends are observed for fat and fiber, as in nixtamalized Quality Protein Maize (NQPM)–soybean tortillas can reach crude fat levels of 10.24% and dietary fiber contents of 3.7%, reflecting the naturally fiber-rich composition of whole pulse flours with an optimized blend of 74.5% NQPM and 25% soybean flour (Hassan et al., 2024). Fortification of nixtamalized maize with common bean flour increases the nutritional and nutraceutical content of tortillas without modifying sensory properties (Treviño-Mejía et al., 2016). In those studies nutritional improvements were obtained by replacing maize up to 30%. Limited information is available on the effects of bean processing on the properties of maize–bean tortillas containing high levels of 30% of bean flour (50:50). This ratio was selected based on preliminary observations, as higher bean proportions produced brittle tortillas with poor rollability.

These studies demonstrate the nutritional potential of maize–pulse blends. However, they have mainly focused on the effect of pulse flour incorporation to nixtamalized maize, without exploring the influence of the nixtamalization processing conditions applied to the pulse component. Therefore, the objective of this study was to evaluate the effects of traditional nixtamalization, ecological nixtamalization, and water cooking applied to four bean types on the rheological, nutritional, physicochemical, textural, and *in vitro* starch hydrolysis properties of tortillas produced from 50:50 blends of nixtamalized maize and treated bean flours.

2. Materials and methods

2.1. Materials

Ayocote bean (*Phaseolus coccineus L.*), black bean (*Phaseolus vulgaris L.*), red bean (*Phaseolus vulgaris var*. "Sangre de toro"), and faba bean (*Vicia faba* L.) were harvested during spring 2024 and bought from a local market in Xonacatlán (State of Mexico, Mexico) with a moisture content ranging from 12 to 14 g/100 g. All samples were stored in polyethylene bags and stored at 4 °C until processing. Food-grade calcium hydroxide ($Ca(OH)_2$; CAS No 1305–62–0**)** and calcium chloride ($CaCl_2$; 97–99% purity; CAS No 10,043-52–4), used for the nixtamalization treatments, were supplied by Alquimia Mexicana (Mexico City, Mexico). Enzymes α-amylase from porcine pancreas, Type VI-B, A3176–2.5MU (9.9 units/mg solid; CAS No 9000–90–2) (Sigma-Aldrich Co, St. Louis, Missouri, USA) and amyloglucosidase from *Aspergillus niger*, 10,113–5 G (120 U/mg; CAS No 9032–08–0) were purchased from Sigma-Aldrich Co. (St. Louis, Missouri, USA).

2.2. Beans and maize flours preparation

Four different types of beans were used: red bean (RB), black bean (BB), ayocote bean (AB) and faba bean (FB). Beans were subjected to three different pretreatments: water-cooked (CK), traditionally nixtamalized (TN), and ecologically nixtamalized (EN). Two batches were prepared for each treatment, and equal portions were mixed to obtain a composite sample. The resulting composite sample was used to prepare the dough and tortilla samples for subsequent analyses. For the water-cooked beans, 500 g each bean was cooked in water at a bean-to-water ratio of 1:2.5 for 2 h at 94 °C. Samples were left to steep for 16 h. After cooking, the grains were drained using a colander until all excess water was removed and washed twice with 0.5 L of water per wash. The grains were drained, milled using a stone grinder (M100, Fumasa, Queretaro, Mexico), dried in a flash dryer (Cinvestav, Queretaro, Mexico) (250 °C inlet air), and re-milled with the 0.5 mm hammer mill screen (Pulvex-200 Mexico City, Mexico).

Nixtamalized bean flour was prepared following the procedures described by Figueroa et al. (2011). For each treatment (traditional or ecological), 500 g of beans were cooked for 2 h at 94 °C in a 1:2.5 bean-to-water ratio with either 1% $Ca(OH)_2$ (TN) or 1% $CaCl_2$ (EN). Samples were left to steep for 16 h. After cooking, the grains were drained using a colander until all excess water was removed and washed twice with 0.5 L of water per wash. The grains were drained, stone-milled, dried in the flash dryer (250 °C inlet air), and ground using the 0.5 mm hammer-mill screen. Samples were stored at 4 °C. All processing conditions, including the bean-to-water ratio, cooking time and temperature, steeping time, washing, draining, and drying conditions, were kept constant across treatments; only the cooking medium differed: water for CK, 1% $Ca(OH)_2$ for TN, and 1% $CaCl_2$ for EN.

For maize, 2 kg of maize were nixtamalized with 1% $Ca(OH)_2$ in 4 L of water (1:2 maize-to-water ratio) and cooked at 94 °C for 30 min. The mixture was then removed from the heat and left to steep for 16 h at room temperature (20 ± 3 °C). After steeping, the nixtamal was drained using a colander until all excess water was removed and washed twice with 0.5 L of water per wash. The grains were then stone-milled, dried in a flash dryer (CINVESTAV, Mexico) at 250 °C for 4 s, and ground using the 0.5 mm hammer-mill screen. Two batches were produced and subsequently pooled together. Samples were stored at 4 °C.

Two independent batches were prepared for each bean–processing treatment and combined in equal proportions to obtain a composite flour sample. Therefore, the two batches were not treated as independent experimental replicates after pooling. Subsequent dough and tortilla samples were prepared from the resulting composite flour sample for the different analyses.

2.3. Rheological analysis of doughs prepared from nixtamalized bean flours

Nixtamalized flours were rehydrated with water at room temperature (14–16 °C) up to 56% moisture. Viscoelastic and textural parameters were evaluated. The viscoelastic properties were evaluated using an AR 2000 rheometer (TA Instruments Inc., New Castle, DE) equipped with parallel steel plates of 40 mm in diameter, at a gap of 2.0 mm. Measurements were performed at 25 °C using a Peltier temperature control system. A strain sweep test was conducted to determine the linear viscoelastic region, applying oscillatory strain from 0.01% to 10% at a constant frequency of 1 Hz. The linear viscoelastic region was identified as the strain range over which G' and G'' remained essentially independent of strain. Based on the strain sweep results, 0.05% strain was selected for the subsequent frequency sweep because both moduli remained stable at this strain. Subsequently, a frequency sweep test was performed in the linear viscoelastic region from 0.1 to 100 Hz at a constant strain of 0.05%. Oscillatory measurements were conducted under continuous oscillation with a sampling time of 3 s and a conditioning time of 3 s. The elastic (G') and viscous (G'') moduli were

recorded as a function of strain and frequency to characterize the viscoelastic behavior of the dough. All rheological parameters were obtained directly from the TA Instruments data acquisition software (version 5.8.2). The analysis allowed evaluation of structural stability and resistance to deformation of the dough under dynamic conditions.

Two independent dough samples were analyzed for each bean–processing treatment combination. Each dough sample was measured in duplicate, and the mean of the duplicate measurements was used for statistical analysis.

Texture profile analysis (TPA) was performed using a TA-XT2 texture analyzer (Texture Technologies Corp., NY. USA). A 35 g sample of dough was placed into a cylindrical mold and compressed with 1 kg to obtain a uniformly compacted sample. The excess of dough was removed using a spatula. The resulting disk had a diameter of 51 mm and a height of 10 mm. Each cylindrical sample was subjected to a two-cycle compression test using a rounded TA-10 probe (1.2 cm diameter) at 40% strain and a test speed of 1 mm/s. The second compression cycle was applied after a 2 s interval. Adhesiveness and cohesiveness were evaluated according to the methodology described by Gasca-Mancera & Casas-Alencáster (2007). Ten independent dough samples were analyzed for each bean–processing treatment combination

2.4. Tortillas preparation

Tortillas were prepared by blending maize and bean doughs in a 50:50 proportion. Portions of dough were used to form 15 cm disks (2 mm thickness) using a manual tortilla press (Casa González, Monterrey, NL, México). The disks were cooked on a griddle preheated to 280 °C for 17 s on the first side, 50 s on the opposite side, and an additional 37 s on the initial side. Cooked tortillas were cooled at ambient temperature to 30 °C. Ten independently prepared tortillas were used for assessing color and texture properties for each bean–-processing treatment combination, and ten additional tortillas were dried at 50 °C for 72 h and combined to obtain a composite tortilla sample for subsequent analyses.

2.5. Proximal composition and mineral content of tortillas

Proximate analysis of the composite tortilla sample was conducted in triplicate using standard AACC International (2010) procedures: moisture (method 44.15.01), ash method (08.12.01), protein (method 46.19.01), fat (method 30.10.01) and crude fiber (method 32.10.01). Results were reported on a dry weight basis.

The concentrations of Na, Mg, Fe, K, *Ca*, and Zn were quantified in the composite tortilla sample by Inductive Coupled Plasma-Mass Spectrometry (ICP-MS) using an Agilent 7850 system (Agilent Technologies, Santa Clara, USA). Prior to analysis, samples were digested by microwave-assisted acid digestion using a Mars 5 Microwave Digestion System (CEM Corporation, Matthews, NC, USA). Mineral contents were expressed on a dry weight basis. The analysis was performed in triplicate as analytical replicates.

2.6. Resistant starch content and in vitro enzymatic hydrolysis of starch by recording viscosity analysis of tortillas

Resistant starch content was determined according to AACC Method 32–40.01, with minor modifications. Briefly, the composite tortilla sample was ground, and dry ground tortilla samples (100 mg) were suspended in 4 mL of 0.1 M sodium maleate buffer (pH 6) containing porcine pancreatic α-amylase (30 U/mL) and amyloglucosidase (3 U/mL) and incubated at 37 °C for 16 h in a shaking incubator (Innova 40, Eppendorf SE, Hamburg, Germany). Following enzymatic hydrolysis, the reaction was terminated by the addition of 4 mL of 99% (v/v) ethanol. Samples were then centrifuged, and the supernatant was discarded. The remaining starch residue was solubilized with 2 mL of 2 M KOH and allowed to stand for 20 min at 4 °C. Subsequently, 8 mL of 1.2 M acetate buffer (pH 3.8) were added, followed by 0.1 mL of amyloglucosidase solution (3300 U/mL). The mixture was incubated at 50 °C for 30 min to complete starch hydrolysis. Subsequently, an aliquot of 20 μL was transferred to a 96-well microplate and mixed with 300 μL of glucose oxidase-peroxidase (GOPOD) reagent, followed by incubation at 50 °C for 20 min. Absorbance was measured at 510 nm against a reagent blank using a microplate spectrophotometer (BioTek Epoch 2, Santa Clara, USA). Resistant starch content was calculated as glucose (mg) × 0.9. The resulting values were then converted to a wet-basis using the sample's moisture content to represent resistant starch per 100 g of sample as consumed. The analysis was performed in triplicate as analytical replicates.

The *in vitro* starch hydrolysis was assessed using a Rapid Visco Analysis RVA4800 (Perkin Elmer, Waltham, Massachusetts, USA) through the digestogram approach, following the method described by Santamaria et al. (2025) with minor modifications. Ground dry tortillas (4.5 g, corresponding to 14 g/100 g moisture basis) were dispersed in 25 mL of distilled water in the RVA canister. The suspension underwent a controlled heating-cooling program consisting of equilibration at 50 °C for 1 min, heating from 50 °C to 95 °C at 10 °C min^{-1}, holding at 95 °C for 2.5 min, and subsequently cooling to 37 °C at the same rate. Stirring was temporarily halted for 36 s to allow the addition of 100 μL of α-amylase solution (450 U/mL). The temperature was then maintained at 37 °C for a further 5 min to monitor enzymatic hydrolysis. The rotational speed was set at 960 rpm for the initial 10 s and subsequently maintained at 160 rpm throughout the analysis, except during enzyme addition, when stirring was stopped. Apparent viscosity was recorded at 1-s intervals.

Data obtained from *in vitro* enzymatic hydrolysis (RVA digestograms) were fitted using OriginPro (OriginLab Corp., Northampton, MA, USA) to the non-linear BoxLucas exponential model (Eq. (1)) as defined by Santamaria et al. (2023).

$$\mu = \mu_{\infty} + (\mu_0 - \mu_{\infty})e^{kt} \quad (1)$$

where μ is the apparent viscosity (cP), μ_o and μ_∞ are the initial and final viscosities, respectively, k (min^{-1}) is the kinetic constant, and t (min) is the hydrolysis time. The *in vitro* starch hydrolysis analysis was performed in triplicate as analytical replicates.

The pH of the gels following enzymatic hydrolysis was determined potentiometrically using a penetration probe (Accumet AB150, Fisher Scientific, Hampton, USA), with measurements performed in triplicate.

2.7. Color and texture analysis of tortillas

Color was measured using a Mini Scan colorimeter (HunterLab, Reston, VA, USA). The CIELab coordinates L^*, a^*, and b^* were recorded, where L^* represents lightness (0 = black, 100 = white), a^* represents redness–greenness, and b^* represents yellowness–blueness.

Texture was determined by a tensile strength test using a Texture Analyzer TA-XT2 (Texture Technologies Corp., Scarsdale, NY, USA). Tensile strength was expressed as the maximum force (N) required to rupture a tortilla strip (8.7 cm long, 3.7 cm wide at each end, and 1.5 cm wide at the narrowest section) mounted between TA-96 tensile grips. The test was performed at a crosshead speed of 2 mm/s according to Calderón-Peralta et al. (2017). Ten tortillas per treatment were analyzed.

2.8. Statistical analysis

Data were analyzed using Minitab® Statistical Software (version 21; Minitab Inc., State College, PA, USA). A general linear model (GLM) was applied to evaluate the effects of bean type (A, BB, FB, and RB) and processing treatment (water cooking, traditional nixtamalization, and ecological nixtamalization) as fixed factors, including their interaction, to evaluate their individual and combined effects. The experimental unit

and number of replicates were defined according to the response variable, as described in Sections 2.3–2.7. When significant differences were detected, means were compared using Tukey's test at $p \leq 0.05$.

3. Results and discussion

3.1. Rheological and textural properties of doughs

The rheological behavior of bean doughs was evaluated through frequency sweep tests, including storage modulus (G), loss modulus (G''), and complex viscosity (η^*).

As shown in Fig. 1-A, both bean type and processing treatment significantly influenced the structural organization and viscoelastic response of the dough systems. All doughs showed an elastic-dominated pattern, with storage modulus G' exceeding the loss modulus G'', indicating a weak gel-like behavior typical of nixtamalized maize systems (Topete-Betancourt et al., 2020). Bean type strongly influenced rheological behavior: red bean-traditional nixtamalization and black bean-traditional nixtamalization presented the highest G' values, suggesting the formation of a more rigid and interconnected elastic network, which could be associated with their chemical composition, especially with protein and starch content. In contrast, FB presented the lowest moduli, indicating softer and less structured systems. As shown in Fig. 1-B, the viscous modulus G'' followed a similar trend to the storage modulus, although with lower values across the frequency range, suggesting stronger intermolecular interactions and greater resistance to deformation compared with EN and CK treatments.

Frequency sweep tests revealed clear shear-thinning behavior, evidenced by the progressive decrease in complex viscosity (η^*) with increasing angular frequency (Fig. 1-C). This pseudoplastic pattern, previously reported for nixtamalized wheat doughs (Rincón-Aguirre et al., 2021), is characteristic of weak gel-like food systems in which physical networks progressively disentangle under oscillatory deformation. In general, traditional nixtamalization produced the highest complex viscosity across the frequency range, followed by EN. CK resulted in the lowest values (Fig. 1-C). The higher viscosity for TN samples is attributed to calcium-starch crosslinking of the gelatinized starch promoted by lime, which enhance flow resistance and structural cohesion (Mendez-Montealvo et al., 2006).

Also, samples presented evidence of the progressive decrease in complex viscosity with increasing angular frequency (Fig. 1-C). This suggests that nixtamalization promotes the development of structured and elastic bean doughs. Nevertheless, previous studies have explained that high viscoelastic moduli may lead to firmer and less extensible tortillas during storage (Hernández-Uribe et al., 2010); therefore, the technological significance of rheological differences observed among treatments should be interpreted together with texture profile analysis to establish their impact on dough handling and final product quality.

From an industrial perspective, these structural features are relevant for improving dough handling, dimensional stability, and rollability during tortilla processing.

In Table 1, the values for adhesiveness and cohesiveness of prepared doughs are presented. These parameters are related to dough handling and machinability for tortilla elaboration, which resulted from dough chemical and structural transformations of starch, proteins, and cell wall components during nixtamalization (Topete-Betancourt et al., 2020). Adhesiveness and cohesiveness were significantly affected by bean type, processing treatment and their interaction ($p < 0.001$). The high F-values obtained for adhesiveness ($F = 3321.33$, 1008.65 and 583.60 for pulse, treatment and pulse x treatment) indicate a strong dependence of surface thickness on both pulse composition and process conditions, whereas cohesiveness showed a significant interaction effect (pulses x treatment: $F = 210.67$). TN increased adhesiveness in most pulse doughs, with the highest effect observed in FBTN and ATN, which reached values above 240 g, which is more adhesive than nixtamalized maize. However, this trend was not observed in black bean doughs,

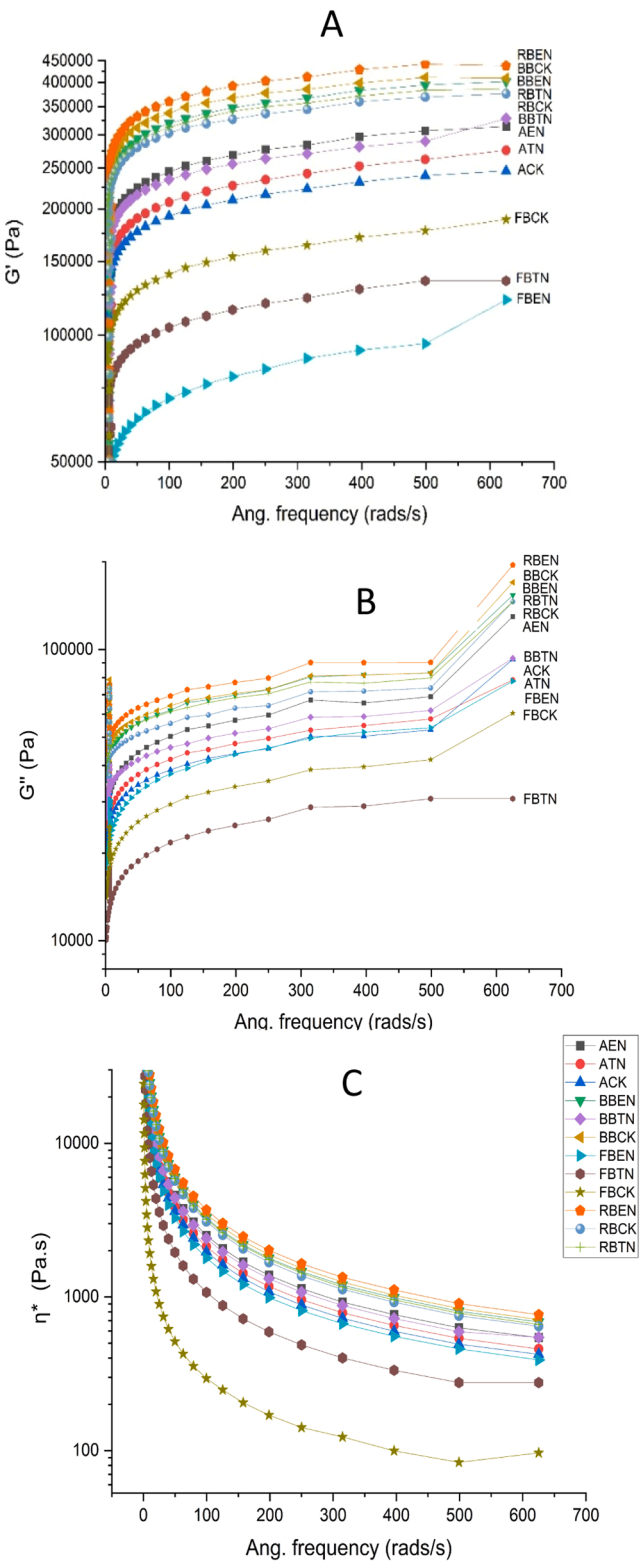


Fig. 1. Effect of bean type and nixtamalization treatment on storage modulus (G') (1-A), loss modulus (G'') (1-B), and complex viscosity (η^*) (1-C).
AEN: Ayocote Ecological Nixtamalization; ATN: Ayocote Traditional Nixtamalization; ACK: Ayocote Cooked; BBEN: Black bean Ecological Nixtamalization; BBTN: Black bean Traditional Nixtamalization; BBCK: Black bean Cooked; FBEN: Faba bean Ecological Nixtamalization; FBTN: Faba bean Traditional Nixtamalization; FBCK: Faba bean Cooked; RBEN: Red bean Ecological Nixtamalization; RBTN: Red bean Traditional Nixtamalization; RBCK: Red bean Cooked.

Table 1
Texture parameters of treated beans doughs.

Sample	Adhesiveness (gf)	Cohesiveness
AEN	211.99 ± 1.57 b	−0.35±0.00 ef
ATN	242.27 ± 1.64 a	−0.29 ± 0.01 c
ACK	169.82 ± 1.57 d	−0.22 ± 0.01 b
BBEN	147.70 ± 2.22 ef	−0.36±0.01 ef
BBTN	143.57 ± 2.15 fg	−0.30± 0.01 c
BBCK	172.73± 2.10 d	−0.32±0.01 cd
FBEN	195.57 ± 3.86 c	−0.16±0.00 a
FBTN	245.30 ± 1.42 a	−0.22 ± 0.01 b
FBCK	174.19 ± 0.80 d	−0.34±0.01 de
RBEN	129.67 ± 2.13 g	−0.21 ± 0.00 b
RBTN	151.08 ± 2.53 e	−0.23 ± 0.00 b
RBCK	138.43 ± 1.47 g	−0.37 ± 0.00 f
MTN	225.85 ± 2.25	−0.34 ± 0.01

Values are means ± SD (n = 10). Different superscript letters within each column indicate significant differences among pulse–treatment combinations (Tukey's test, $p < 0.05$).
AEN: Ayocote Ecological Nixtamalization; ATN: Ayocote Traditional Nixtamalization; ACK: Ayocote Cooked; BBEN: Black bean Ecological Nixtamalization; BBTN: Black bean Traditional Nixtamalization; BBCK: Black bean Cooked; FBEN: Faba bean Ecological Nixtamalization; FBTN: Faba bean Traditional Nixtamalization; FBCK: Faba bean Cooked; RBEN: Red bean Ecological Nixtamalization; RBTN: Red bean Traditional Nixtamalization; RBCK: Red bean Cooked; MTN: Maize traditional nixtamalization.

where CK showed higher adhesiveness values than TN. This response closely parallels the rheological behavior, where TN presented the highest complexity viscosity (η^*) and elastic moduli (G).

The high alkalinity of the TN induces hydrolysis of the maize pericarp, releasing arabinoxylans that act as hydrocolloids or gums, which provide desirable cohesiveness and adhesiveness (Topete-Betancourt et al., 2020). Although part of these cell wall components may be lost in the nejayote, during alkaline cooking and washing, the remaining arabinoxylans may undergo structural modifications that increase their solubility and functionality within the dough matrix, enhancing their hydrocolloid behavior (Sunico et al., 2021).

The use of alternative calcium salts has been reported to lower alkalinity and not hydrolyze the pericarp completely, producing a weaker and less elastic structure compared to TN (Santiago-Ramos, Figueroa-Cárdenas, and Véles-Medina, 2018). These results confirm that textural responses are directly linked to the viscoelastic structure. From an industrial standpoint, controlling adhesiveness and cohesiveness through nixtamalization conditions is critical to ensure dough machinability, reduce sticking during sheeting, and maintain consistent tortilla quality in legume-fortified formulations.

When compared to commercial nixtamalized maize flour (MTN) (adhesiveness 225.85 and cohesiveness −0.34), RB doughs exhibited the closest functional balance, combining sufficiently low adhesiveness and avoiding processing issues with adequate cohesiveness and superior elastic moduli, supporting their suitability for tortilla production. This functional balance enables the incorporation of nutritionally enriched legume ingredients without compromising process efficiency or final product quality.

3.2. Proximal composition, resistant starch, and mineral content of tortillas

Table 2 summarizes the proximate and mineral composition of maize-bean tortillas and highlights the differentiated contribution of bean type and nixtamalization treatment to macronutrient and mineral content. Protein content was mainly driven by bean type, as evidenced by the strong GLM (Generalized linear model) effect (F = 604.34,

Table 2
Proximate analysis and mineral composition of nixtamalized and bean-fortified tortillas, expressed on dry weight.

Proximal analysis						
Sample	Protein (g/100 g)	Crude Fiber(g/100 g)	Total Fat (g/100 g)	Carbohydrates (g/100 g)	Ashes (g/100 g)	Energy (kcal/100 g)
AEN	16.54±0.06cd	3.31±0.04abc	3.12±0.06ab	73.86±0.08b	3.15±0.01cd	396.34±0.41cd
ATN	16.63±0.08c	3.26±0.01bcd	3.06±0.03ab	73.53±0.10b	3.50±0.02a	394.80±0.13e
ACK	16.11±0.19de	3.36±0.06ab	3.19±0.06a	74.73±0.24ab	2.58±0.03h	398.92±0.47a
BBEN	15.31±0.22f	2.95±0.00e	2.86±0.01bc	75.83±0.24ab	3.03±0.04e	396.29±0.20cd
BBTN	15.75±0.07ef	2.78±0.01f	2.78±0.07c	75.38±0.04ab	3.29±0.01b	395.16±0.38de
BBCK	15.78±0.13e	3.15±0.03d	3.00±0.00abc	75.35±0.14ab	2.70±0.00 g	397.89±0.08ab
FBEN	18.11±0.05a	3.18±0.01cd	2.99±0.09abc	72.91±0.07b	2.79±0.00f	397.44±0.46bc
FBTN	17.50±0.35b	3.13±0.01d	2.79±0.17c	73.50±0.51b	3.07±0.00e	395.40±0.86de
FBCK	17.58±0.36b	3.23±0.04cd	3.08±0.14ab	73.55±0.35b	2.53±0.00h	398.78±0.71ab
RBEN	14.68±0.05 g	3.20±0.12cd	2.97±0.01abc	79.59±7.34a	3.20±0.01c	395.65±0.37de
RBTN	14.61±0.02 g	2.92±0.07e	2.90±0.22bc	76.42±0.24ab	3.14±0.04d	396.08±1.13cde
RBCK	14.65±0.04 g	3.39±0.06a	3.05±0.07ab	76.24±0.04ab	2.64±0.00 g	397.91±0.46ab
MTN	9.08±0.01	2.99±0.15	2.38±0.23	83.90±0.15	3.20±0.01	399.40±1.46
Mineral content mg/100 g						
	Na	Mg	Fe	K	Ca	Zn
AEN	72.08±3.49d	880.29±5.16b	21.67±0.14cde	5952.40±50.20bc	2923.52±3.550 g	23.87±0.51ab
ATN	61.23±4.30ef	906.19±0.54a	23.00±0.11abc	6213.40±5.91a	3994.92±9.95b	25.50±0.89a
ACK	57.31±1.27fg	814.54±3.89d	22.25±1.51bcd	6050.33±54.55b	772.795±4.387k	22.57±1.09bc
BBEN	61.96±0.98ef	830.46±6.92c	23.65±0.46ab	5367.13±14.66e	3369.92±7.938e	17.27±0.69f
BBTN	62.53±1.18e	886.06±3.57b	22.85±0.81abc	5830.73±20.72d	3777.57±0.663c	17.65±0.83ef
BBCK	57.48±1.58fg	809.61±7.85d	24.20±0.51a	5855.88±120.60cd	1034.69±4.3878j	19.35±0.77de
FBEN	79.63±0.60c	607.29±5.97i	23.67±0.14ab	4634.90±44.36h	3303.44±12.99f	20.90±1.01cd
FBTN	87.58±0.49b	640.96±3.86 g	20.00±0.23ef	4826.15±33.28f	4088.22±14.76a	20.05±0.37d
FBCK	95.08±1.93a	624.61±10.0h	21.72±0.83cde	5126.10±13.42f	1202.87±6.899i	22.07±0.86bc
RBEN	47.36±0.34i	682.49±1.47f	19.42±0.89f	5116.45±11.63f	2727.77±9.670h	19.45±0.31de
RBTN	54.38±0.54gh	758.46±2.94e	21.37±0.72cde	5906.38±49.53cd	3626.14±11.37d	20.77±0.92cd
RBCK	50.48±1.81hi	687.46±3.29f	20.65±0.75def	5910.78±6.11cd	688.195±15.5l	17.00±0.83f
MTN	9.66±0.11	131.80±0.29	2.49±0.17	237.13±31.55	343.33±0.365	1.99±0.16

Values are means ± SD (n = 3). Different superscript letters within each column indicate significant differences among pulse–treatment combinations (Tukey's test, $p < 0.05$).
AEN: Ayocote Ecological Nixtamalization; ATN: Ayocote Traditional Nixtamalization; ACK: Ayocote Cooked; BBEN: Black bean Ecological Nixtamalization; BBTN: Black bean Traditional Nixtamalization; BBCK: Black bean Cooked; FBEN: Faba bean Ecological Nixtamalization; FBTN: Faba bean Traditional Nixtamalization; FBCK: Faba bean Cooked; RBEN: Red bean Ecological Nixtamalization; RBTN: Red bean Traditional Nixtamalization; RBCK: Red bean Cooked; MTN: Maize traditional nixtamalization.

$p \leq 0.001$), whereas the effect of processing was not significant. However, a significant bean x treatment interaction ($F = 9.88$, $p \leq 0.001$) indicates that protein retention during nixtamalization is bean-dependent. This trend corroborates earlier findings suggesting that the impact of alkaline processing on protein solubility and matrix organization varies according to bean source (Santiago-Ramos et al., 2018). Faba bean-maize tortillas exhibited the highest protein levels (17.5–18.1%), reflecting their high intrinsic protein content and confirming bean selection as the main driver of protein enhancement. The protein content obtained falls within the range reported for cereal – bean systems, comparable to tortillas with maize and extruded chickpea flour and maize chickpea tortillas (13–13.8%) (Bon-Padilla et al., 2022), while higher levels are generally associated with elevated substitution rates (15–30%) of protein-rich beans such as lentils or soy (17–22%) (Cortes et al., 2023; Sánchez-Villa et al., 2020).

Crude fiber content was significantly influenced by treatment, bean type and their interaction, indicating that fiber retention depends on both processing and bean matrix characteristics. In general, traditional nixtamalization resulted in the lowest fiber values, consistent with alkaline hydrolysis and partial removal of pericarp (Campechano Carrera et al., 2012). In contrast, EN showed intermediate retention while CK achieved the highest crude fiber content, suggesting improved preservation of cell wall polysaccharides when only cooking was applied. In general, beans incorporation resulted in comparable or slightly higher fiber contents in several formulations, of the tortillas, which might be expected due to their higher fiber content (15–39%) (Alcázar-Valle et al., 2021) compared to maize. Ayocote bean provided the highest fiber content, likely due to differences in seed coat composition and resistance to thermal modification. Maintaining higher fiber levels is relevant for developing tortillas with improved physiological functionality while preserving acceptable processing behavior. Fat content was influenced by treatment and beans type, but not by their interaction. Beans increased the fat content of the tortillas, compared to the maize tortilla. Regarding treatments, CK retained the highest lipid values, followed by EN and TN, consistent with progressively lipid losses during alkaline processing through saponification of triglycerides and the removal of lipid rich germ and pericarp during washing (Hernández-Salazar et al., 2006; Santiago-Ramos, et al., 2018). Tortillas with ayocote bean displayed higher fat levels, in agreement with the reported lipid content of ayocote bean (Mariscal-Moreno et al., 2021). These compositional shifts enable modulation of energy density and macronutrient balance according to specific nutritional targets.

Nixtamalization and pulse fortification significantly modified the ash content and mineral composition (Na, Mg, Fe, K, *Ca* and Zn) of tortillas (Table 2). GLM revealed that bean type and processing treatment, and their interactions (pulse x nixtamalization) had significant effects ($p \leq 0.001$) on most minerals, confirming their joint impact on the final mineral profile. Nevertheless, the strongest effect was observed for the nixtamalization process used while bean incorporation exerted a secondary effect. Total ash ranged from 2.53 to 3.50 g/100 g with the lowest values observed in CK samples, which even contained lower ash content than the maize tortilla. Nixtamalization led to significantly higher mineral content due to the alkali treatment. Particularly, there was a significant increase in calcium due to *Ca* uptake from the alkaline cooking medium (Hernández Salazar et al., 2006; Mariscal-Moreno et al., 2021). Thus, the higher calcium concentrations observed in nixtamalized tortillas primarily reflect calcium introduced during processing, whereas the contribution of endogenous minerals from the pulses varied according to bean type and mineral. The higher ash content reflects the contribution of minerals incorporated from nixtamalization, while the incorporation of endogenous minerals from pulses did not consistently increase ash levels compared with maize tortillas. Only the ayocote traditional nixtamalization showed higher ash content. TN treatments exhibited *Ca* concentrations up to fivefold higher than the maize tortilla control (MTN), with the highest values observed in TN tortillas regardless of bean type, while CK samples consistently showed the lowest *Ca* levels. EN treatments using $CaCl_2$ also increased *Ca* concentrations compared with CK samples, although values remained lower than those obtained with TN. This difference is mainly attributed to the lower alkalinity of $CaCl_2$ systems, which preserve the pericarp and cell wall structures and limit Ca^{2+} diffusion into the matrix compared with the strong structural disruption caused by $Ca(OH)_2$ during traditional nixtamalization (Rincón-Aguirre et al., 2021; Santiago-Ramos et al., 2018; Topete-Betancourt, 2020). This agrees with previous works demonstrating that $Ca(OH)_2$ nixtamalization (traditional nixtamalization) acts as an effective calcium enrichment process, increasing *Ca* content, resulting in a 5- to 16-fold increase in calcium content compared with raw maize (Chuck Hernández and Serna-Saldivar, 2019). This substantial calcium enrichment resulting from the nixtamalization process has important implications for populations with limited access to dairy products, positioning nixtamalized maize–bean tortillas as a potential dietary calcium source (Acevedo-Martinez and de Mejia, 2021).

Iron and zinc contents were significantly influenced by bean type and bean x nixtamalization interaction, whereas the main effect of nixtamalization alone was not significant for iron. Nixtamalization largely preserves Fe and Zn, while bean incorporation enhances their concentration. Pulses such as soybean and chickpea are recognized as a source of these micronutrients, and their inclusion in nixtamalized maize have improved overall mineral content (Bon-Padilla et al., 2022; Hassan et al., 2024). The high retention of Fe and Zn during nixtamalization (>90%) reported in mechanistic studies explains the stability of these minerals despite pericarp removal (Gallego-Castillo et al., 2021). Fortification of tortillas represents an effective strategy to improve the nutritional status of populations that regularly consume cereal-based staple foods worldwide. Given their high per capita intake in many regions, tortillas provide an ideal vehicle for increasing dietary mineral intake, particularly iron, which is essential for preventing iron-deficiency anemia and supporting metabolic and cognitive function across age groups. Also, sodium content was affected by bean type and its interaction with nixtamalization process, with FB formulations showing the highest Na levels and the lowest. (Acosta-Estrada et al., 2023). In contrast, potassium and magnesium decreased in some treatments, particularly in ecological nixtamalization samples, likely due to leaching during cooking and washing, as both minerals are highly soluble (Mariscal-Moreno et al., 2021). However, TN and specific bean combinations (ATN and BBTN) offset these losses, consistent with findings that mineral rich alkali or ashes can mitigate Mg and K depletion (Odukoya et al., 2022; Santiago-Ramos et al., 2018). Overall, combining traditional nixtamalization with bean fortification resulted in tortillas with an improved mineral profile, although low reductions were observed in K and Mg, but these are nutritionally less critical.

3.3. Resistant starch and in vitro starch hydrolysis of tortillas

The effects of treatment and bean type on pH, kinetic *in vitro* hydrolysis parameters (μ_0, k, μ_∞), and resistant starch (RS) content are presented in Table 3. Resistant starch content was significantly affected by treatment ($p \leq 0.05$), whereas no overall bean effect was detected. $CaCl_2$ treatments generally resulted in the highest RS values, particularly in faba bean tortillas (4.82–5.04 g/100 g). In contrast, $Ca(OH)_2$ treatments tended to yield lower RS contents in faba bean, red bean, and black bean tortillas (3.78, 4.38 and 3.89 g/100 g, respectively), with the latter showing a significantly ($p \leq 0.05$) lower value. A different pattern was reported by Santiago-Ramos et al. (2018), who observed a marked reduction in resistant starch content from 10.10% in raw black bean flours to 4.19% following $Ca(OH)_2$- and $CaCl_2$-based nixtamalization. This apparent discrepancy can be attributed to differences in processing sequence and matrix state, as nixtamalization alone involves intense hydration, alkaline treatment, and thermal input that promote starch gelatinization and disruption of native resistant structures. Conversely, subsequent tortilla processing, including shaping and cooking, may

Table 3
pH, *in vitro* enzymatic hydrolysis parameters and resistant starch content of tortillas.

Sample	pH				μ_0 (cP)				k (min^{-1})				μ_∞ (cP)				RS (g/100 g w.b.)			
AEN	6.76	±	0.01	a	459	±	8	a	1.24	±	0	e	85	±	0	a	4.82	±	0.07	d
ATN	7.5	±	0.01	g	468	±	19	a	0.73	±	0.07	bc	121	±	4	c	4.77	±	0.07	cd
ACK	6.9	±	0.01	b	620	±	12	bc	0.92	±	0.06	cd	143	±	0	d	3.92	±	0.07	b
BBEN	6.79	±	0	ab	673	±	7	c	1.4	±	0.01	e	159	±	5	e	4.78	±	0.18	cd
BBTN	7.63	±	0.01	h	578	±	20	b	0.62	±	0.03	ab	163	±	6	e	3.89	±	0.18	b
BBCK	6.99	±	0	c	778	±	15	d	1.07	±	0.01	d	200	±	4	g	4.79	±	0.22	cd
FBEN	6.8	±	0.01	b	438	±	5	a	0.99	±	0.04	d	92	±	5	a	5.04	±	0.04	d
FBTN	7.36	±	0	f	455	±	10	a	0.92	±	0.03	cd	105	±	2	b	3.78	±	0.09	bc
FBCK	6.89	±	0.01	c	489	±	10	a	0.81	±	0	bc	117	±	2	bc	3.45	±	0.11	b
RBEN	6.89	±	0.01	c	864	±	25	e	1.37	±	0.07	e	228	±	1	h	4.38	±	0.1	c
RBTN	7.62	±	0.01	h	588	±	7	b	0.66	±	0.03	ab	187	±	1	f	4.38	±	0.15	c
RBCK	7.08	±	0.02	e	802	±	28	d	1	±	0.03	e	227	±	0	h	4.89	±	0.14	d
MTN	7.92	±	0.03		2265	±	32		0.43	±	0.01		1065	±	27		2.79	±	0.07	
p-value																				
Treatment	0	0.0006	0	0.0007	0.0242															
Bean	0	0	-	0	-															

Means with different letters within a row were significantly different ($p \leq 0.05$) (n = 3).
AEN: Ayocote Ecological Nixtamalization; ATN: Ayocote Traditional Nixtamalization; ACK: Ayocote Cooked; BBEN: Black bean Ecological Nixtamalization; BBTN: Black bean Traditional Nixtamalization; BBCK: Black bean Cooked; FBEN: Faba bean Ecological Nixtamalization; FBTN: Faba bean Traditional Nixtamalization; FBCK: Faba bean Cooked; RBEN: Red bean Ecological Nixtamalization; RBTN: Red bean Traditional Nixtamalization; RBCK: Red bean Cooked; MTN: Maize traditional nixtamalization.

favor starch retrogradation during cooling and storage, thereby promoting RS3 formation. These observations suggest that thermal processing and post-processing conditions, instead of nixtamalization per se, are critical determinants of final resistant starch levels in bean-based tortillas. Consistent with this interpretation, the present results suggest that calcium-mediated ionic interactions may contribute to the differences in resistant starch formation observed among treatments. These interactions may promote molecular reorganization and stabilization of starch structures, as well as interactions between starch and other matrix components (proteins, lipids, and dietary fiber), potentially reducing enzymatic accessibility and susceptibility to hydrolysis. Although these interactions were not directly characterized in the present study, this interpretation is consistent with the observed differences in resistant starch content and hydrolysis behavior and with previously reported effects of calcium on starch structural organization (Bello-Perez et al., 2020). Similar effects have been reported for tortillas made with nixtamalized white maize, where calcium salts (lime, calcium carbonate and calcium propionate) induced structural rearrangements, including enhanced amylose-amylose associations and increased formation of ordered or crystalline domains, thereby increasing resistance to α-amylase hydrolysis (Mariscal Moreno et al., 2015). By contrast, strongly alkaline conditions associated with $Ca(OH)_2$ treatment may intensify starch gelatinization and partial depolymerization, limiting the ability of starch chains to reassociate into stable retrograded structures, which are primarily responsible for RS3 formation (Bello-Perez et al., 2020). Relative to corn (2.79 g/100 g), all bean tortillas showed higher RS values, in line with the higher amylose content and distinct granular and molecular features of legume starches, which favor the formation of enzyme-resistant structures (Santiago-Ramos et al., 2018).

The results for *in vitro* starch hydrolysis demonstrated that treatment exerted a statistically significant effect on all measured parameters ($p \leq 0.0242$), including pH, initial viscosity (μ_0), kinetic constant (k), final viscosity (μ_∞), and RS content, highlighting the central role of nixtamalization severity in affecting starch accessibility and hydrolysis behavior. In contrast, bean type significantly affected pH, μ_0, and μ_∞ ($p < 0.0001$), whereas no overall bean effect was detected for the kinetic constant k and RS content, indicating that these latter responses were predominantly governed by treatment conditions rather than botanical origin, consistent with the strong physicochemical impact of alkaline–thermal processing on starch gelatinization and granule disruption. The maize tortillas, although exhibiting markedly higher viscosities and lower RS content than all bean-derived samples, were not included in the statistical analysis and therefore served exclusively as a qualitative benchmark for comparison, denoting the starch dilution and structural shielding effects associated with legume incorporation.

The pH of the slurries varied significantly across treatments ($p < 0.001$). In all cases, $Ca(OH)_2$ increased the pH relative to the corresponding water and $CaCl_2$-treated samples, reflecting the strong alkalinity of calcium hydroxide and its capacity to elevate medium pH, in agreement with previous reports on calcium salts in nixtamalized systems (Mariscal Moreno et al., 2015). Conversely, $CaCl_2$ consistently yielded the lowest pH values, which can be attributed to calcium's hydrolytic behavior and formation of slightly acidic aqueous equilibria. Across bean types, pH values ranged from 6.76 (Ayocote $CaCl_2$) to 7.63 (Black bean $Ca(OH)_2$), a particularly relevant range because α-amylase activity is favored at near-neutral pH, thereby modulating hydrolysis kinetics.

Untreated (water-based) samples exhibited moderate to high initial viscosities (489–802 cP), depending on bean type, whereas both $Ca(OH)_2$ and $CaCl_2$ treatments markedly reduced μ_0 ($p < 0.001$), except for $CaCl_2$-processed red bean tortillas. The lowest μ_0 values were observed in Ayocote and Faba beans treated with $Ca(OH)_2$ or $CaCl_2$, ranging from 438 to 468 cP, with the reduction being significant only in Ayocote beans. This trend showed the calcium-induced weakening of interparticle and matrix cohesion during initial shear, possibly through competitive ion binding to anionic sites on proteins or polysaccharides (Jane, 1993).

The kinetic constant k quantifies the rate at which apparent viscosity decays under *in vitro* enzymatic hydrolysis and constant shear, serving as an index of structural susceptibility to α-amylase. $CaCl_2$ treatment yielded the highest k values in each bean (red bean 1.37 min^{-1}; black bean 1.40 min^{-1}; ayocote bean 1.24 min^{-1}; faba bean 0.99 min^{-1}), indicating rapid viscosity loss and accelerated hydrolysis of the starch. Water and $Ca(OH)_2$ treatments produced slower starch hydrolysis, typically in the range from 0.62 to 1.07 min^{-1}. $Ca(OH)_2$ frequently lowered the hydrolysis kinetic (except for faba bean) relative to water within a given bean (red bean: from 1.00 to 0.66; black bean: from 1.07 to 0.62). In contrast, $CaCl_2$ were characterized by faster hydrolysis kinetics, implying that Ca^{2+} ions in a neutral to slightly acidic environment provided more favorable conditions for pancreatic α-amylase activity. This interpretation is further supported by the strong negative correlation observed between pH and the kinetic constant k ($r = -0.85$, $p \leq 0.05$), indicating that increasing pH is associated with a marked reduction in hydrolysis rate. The corn control, included solely as a

reference, exhibited a substantially lower k value (0.43 min^{-1}).

The parameter μ_∞ represents the infinite-time apparent viscosity predicted by the Box-Lucas-type model. For all beans, except for $CaCl_2$-treated red bean and faba bean tortillas treated with $Ca(OH)_2$, μ_∞ decreased with calcium treatments relative to water, with $CaCl_2$ generally producing the lowest final viscosities. The largest decreases in μ_∞ relative to water occurred in ayocote and faba beans treated with $CaCl_2$, where the value dropped to 85–92 cP. In red bean and faba bean, μ_∞ was less sensitive to $CaCl_2$, suggesting greater resistance to complete viscosity loss despite rapid initial hydrolysis kinetics. Across beans and treatments, μ_∞ values for beans remained substantially below the corn control (1065 cP).

3.4. Color and texture analysis of tortillas

Fig. 2 summarizes the combined effects of bean type and processing treatment on tortilla color parameters (L^*, a^* and b^*) (Fig. 2-A) and hardness (Fig. 2-B). In all cases, the statistical analysis demonstrated highly significant ($p < 0.001$) main effects of bean and treatment, as well as strong bean x nixtamalization interactions, indicating that the impact of alkaline processing on color and texture depends on the type of bean incorporated.

Lightness was mainly driven by processing conditions, with EN treatments showing the highest values and TN samples being darker due to the yellowing associated with lime-based nixtamalization (Escalante-Aburto et al., 2020). Bean type further modulated this effect; tortillas with FB exhibited the greatest luminosity, whereas BB samples were darker. The significant bean x nixtamalization interactions indicate that pulses modify the optical response to nixtamalization. Relative to maize tortillas ($L^* = 61.82 \pm 0.12$), all nixtamalized bean-maize tortillas were darker, particularly TN treatments. From a technological perspective, these color shifts are relevant for defining acceptable formulation limits in pulse-fortified tortillas; additional sensorial analyses are needed in future work.

The a^* values were also affected by both factors. TN treatments, especially RBTN and ATN showed positive a^* values, indicating a shift toward red color driven by pH- induced pigmented transformations and protein-phenolic interactions (Acevedo-Martinez and de Mejia, 2021). For yellowness (b^*), a similar pattern to L^* values were obtained specially for EN and CK treatments were the samples with higher values. These results agree with reports that nixtamalization enhances yellow coloration through flavonoid exposure and Maillard reactions while pulses contribute additional pigments and modify pH, intensifying color development (Acevedo-Martinez and de Mejia, 2021; Escalante-Aburto et al., 2020). Compared with maize flour tortillas ($b^* = 19.92 \pm 0.07$), nixtamalized bean-maize tortillas maintained comparable or higher yellowness, indicating that fortification preserves the expected visual identity. The relatively familiar color profile suggests that the visual changes associated with bean incorporation may not necessarily preclude the potential application of nutritionally enhanced tortillas; however, consumer acceptance should be confirmed through sensory evaluation.

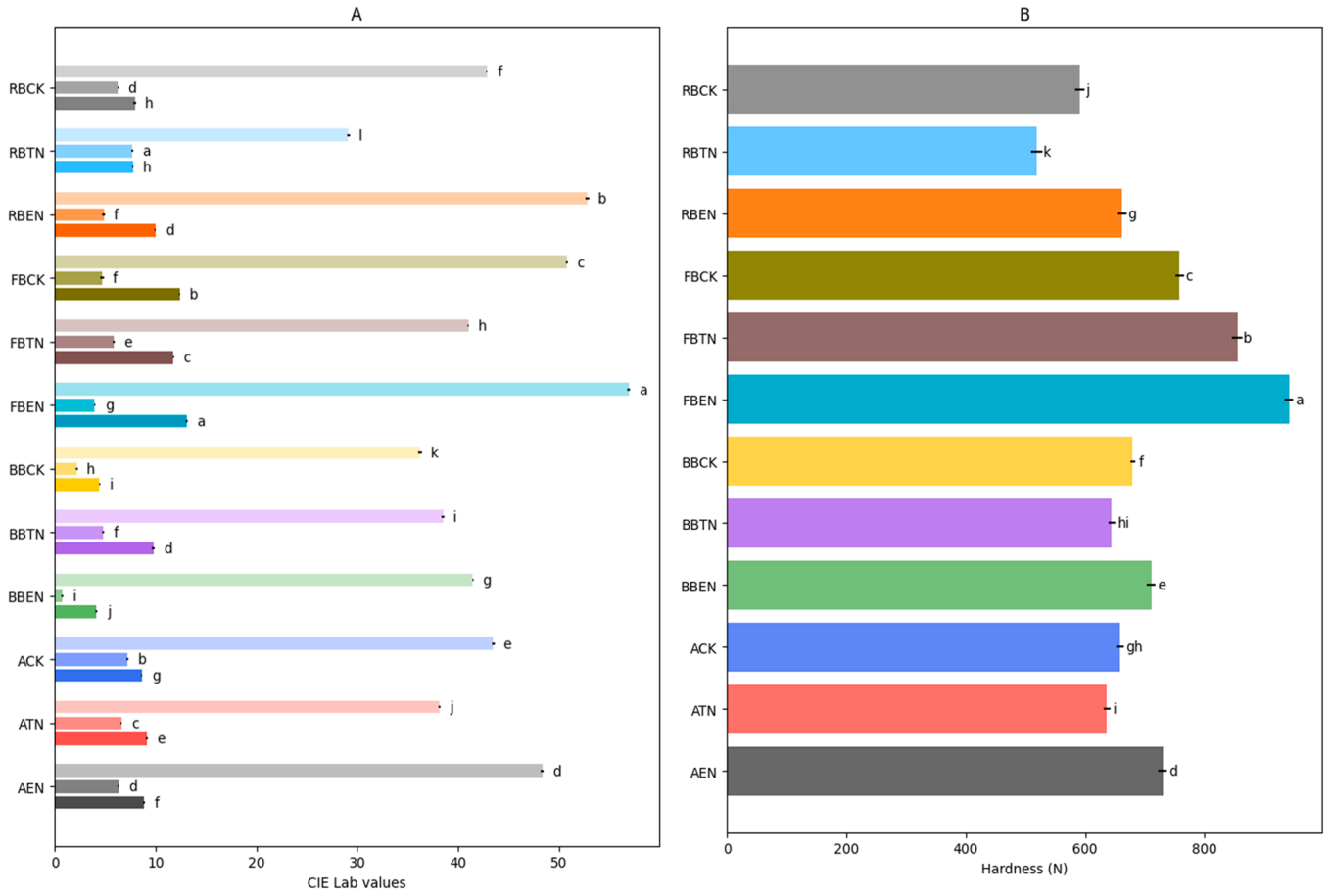


Fig. 2. Color characteristics and hardness of tortillas. (A) CIE L^*, a^*, and b^* values. (B) Hardness (N). Values are expressed as mean ± SD (n = 10). Different letters indicate significant differences ($p < 0.05$).
AEN: Ayocote Ecological Nixtamalization; ATN: Ayocote Traditional Nixtamalization; ACK: Ayocote Cooked; BBEN: Black bean Ecological Nixtamalization; BBTN: Black bean Traditional Nixtamalization; BBCK: Black bean Cooked; FBEN: Faba bean Ecological Nixtamalization; FBTN: Faba bean Traditional Nixtamalization; FBCK: Faba bean Cooked; RBEN: Red bean Ecological Nixtamalization; RBTN: Red bean Traditional Nixtamalization; RBCK: Red bean Cooked; MTN: Maize traditional nixtamalization.

Hardness was significantly influenced by beans and treatment ($p < 0.001$), with pulse type exerting the strongest effect. FB tortillas were the hardest, followed by BB and A, whereas RB produced the softest tortillas. EN treatments yielded higher hardness values than CK and TN, suggesting that milder alkaline conditions favor the structural matrix.

The significant bean x nixtamalization interaction revealed distinct technological behaviors: FBEN exhibited the highest hardness, while RBTN tortillas were the softest. Also tortillas with EN presented higher crude fiber values, which can increase tortilla hardness. From a technological standpoint, controlling hardness through bean selection and nixtamalization strategy is essential to optimize rollability, foldability, and overall textural quality during tortilla processing.

From a mechanistic perspective, these trends reflect the interplay between starch gelatinization, calcium-mediated crosslinking, and the disruptive effects of bean proteins and fibers on the starch network. While legume incorporation generally softens tortillas, FB formulations appear to counterbalance this effect, likely due to higher protein content and stronger protein–starch interactions, consistent with previous reports on high-protein systems (Hassan et al., 2024; Sánchez-Villa et al., 2020). These structure–texture relationships provide a framework for designing pulse-fortified tortillas with targeted sensory and functional attributes.

The novelty of this study resides in showing that nutritional and technological enhancement of maize tortillas can be achieved through coordinated control of nixtamalization intensity and bean selection, without increasing energy density. By comparing traditional, ecological, and control treatments, we demonstrate that calcium enrichment is process-driven, whereas protein, fiber, carbohydrate reduction, and energy redistribution depend mainly on bean type, while iron and zinc remained stable under alkaline conditions. This formulation-based strategy operates within conventional processing schemes and avoids additional unit operations.

4. Conclusion

The results confirm that bean fortification integrated with nixtamalization effectively enhances the nutritional quality of maize tortillas. While nixtamalization remained the dominant factor for calcium enrichment, bean selection primarily governed protein, fiber, carbohydrate, and energy profiles. Compared with conventional maize tortillas (9.08% protein, 2.99% crude fiber), bean-fortified formulations achieved higher protein levels (up to ~18%) and comparable or slightly increased fiber contents, alongside a reduction in total carbohydrates of up to 10 percentage units depending on bean type. Energy values remained within a narrow range (≈395–399 kcal/100 g), indicating that nutritional improvements were achieved through macronutrient redistribution rather than caloric dilution. Despite moderate losses of potassium and magnesium during alkaline processing, mineral stability was preserved in specific bean–treatment combinations. *In vitro* starch hydrolysis and resistant starch formation in pulse-based tortillas were influenced primarily by calcium chemistry rather than botanical origin, with $CaCl_2$ favoring higher resistant starch levels and promoting faster hydrolysis, whereas $Ca(OH)_2$ was associated with slower starch hydrolysis.

These findings identify calcium salt selection and post-processing conditions as critical levers for modulating glycemic-relevant carbohydrate fractions in legume-fortified tortillas. These findings support the potential of bean-fortified nixtamalized tortillas as a strategy for improving the nutritional profile of a staple food while maintaining relevant technological properties. However, sensory evaluation was not performed in this study, and further research is needed to assess consumer acceptance of the developed tortillas. Collectively, these results provide a basis for further investigation of maize–bean nixtamalized systems.

Ethics statement

This study did not involve human participants, human biological materials, or live animals. Therefore, ethical approval and informed consent were not required.

CRediT authorship contribution statement

Rosa María Mariscal-Moreno: Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Juan de Dios Figueroa-Cárdenas:** Methodology, Funding acquisition, Formal analysis. **Cristina M. Rosell:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Nicola Gasparre:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **José Juan Véles-Medina:** Methodology, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Cristina M. Rosell reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. Cristina M. Rosell reports financial support was provided by Canada Foundation for Innovation. Cristina M. Rosell reports financial support was provided by Research Manitoba. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Funding from the Discovery Grant (RGPIN-2023-04289) of the Natural Sciences and Engineering Council (NSERC), Canada Foundation for Innovation (44211) and Research Manitoba (6399).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.fufo.2026.101237.

Data availability

Data will be made available on request.

References

Acevedo-Martinez, K.A., De Mejia, E.G., 2021. Fortification of maize tortilla with an optimized chickpea hydrolysate and its effect on dppiv inhibition capacity and physicochemical characteristics. Foods 10, 1835. https://doi.org/10.3390/foods10081835.

Acosta-Estrada, B.A., Serna-Saldívar, S.O., Chuck-Hernández, C., 2023. Nutritional assessment of nixtamalized maize tortillas produced from dry masa flour, landraces, and high yield hybrids and varieties. Front. Nutr. 10. https://doi.org/10.3389/fnut.2023.1183935.

Alcázar-Valle, M., García-Morales, S., Mojica, L., Morales-Hernández, N., Sánchez-Osorio, E., Flores-López, L., Enríquez-Vara, J.N., Lugo-Cervantes, E., 2021. Nutritional, antinutritional compounds and nutraceutical significance of native bean species (*Phaseolus spp.*) of Mexican cultivars. Agriculture 11, 1031. https://doi.org/10.3390/agriculture11111031.

Bello-Perez, L.A., Flores-Silva, P.C., Agama-Acevedo, E., Tovar, J., 2020. Starch digestibility: past, present, and future. J. Sci. Food Agric. 100, 5009–5016. https://doi.org/10.1002/jsfa.8955.

Bon-Padilla, B.K., Reyes-Moreno, C., Milán-Carrillo, J., Reynoso-Camacho, R., Gómez-Aldapa, C.A., Gómez-Favela, M.A., Gutiérrez-Dorado, R., 2022. Tortillas made from nixtamalized maize and extruded chickpea flours: a product with improved *in vitro* nutritional and antihypertensive properties. Cereal Chem. 99, 1154–1165. https://doi.org/10.1002/cche.10577.

Boukid, F., Zannini, E., Carini, E., Vittadini, E., 2019. Pulses for bread fortification: a necessity or a choice? Trends Food Sci. Technol. 88, 416–428. https://doi.org/10.1016/j.tifs.2019.04.007.

Calderón-Peralta, C.V., Jiménez-Hernández, J., Maldonado-Astudillo, Y.I., Flores-Casamayor, V., Arámbula-Villa, G., Salazar, R., 2017. Influence of Hymenaea

courbaril gum as a new additive on nixtamalized flour properties and quality of tortilla. J. Food Process Eng. 40, e12525.
Campechano Carrera, E.M., de Dios Figueroa Cárdenas, J., Arámbula Villa, G., Martínez Flores, H.E., Jiménez Sandoval, S.J., Luna Bárcenas, J.G., 2012. New ecological nixtamalisation process for tortilla production and its impact on the chemical properties of whole corn flour and wastewater effluents. Int. J. Food Sci. Technol. 47, 564–571. https://doi.org/10.1111/j.1365-2621.2011.02878.x.
Chuck Hernández, C.E., Serna-Saldivar, S.O., 2019. Soybean-fortified nixtamalized corn tortillas and related products. Flour Breads Their Fortif. Health Dis. Prev. 319–332. https://doi.org/10.1016/B978-0-12-814639-2.00025-3.
Cortes, N.G., Vera, R.J., Jiménez, A.L.L., Gálvez, A.B., Ramírez, M.O., 2023. Development of a more nutritious tortilla in Mexico. Int. J. Health Sci. 3, 2–12. https://doi.org/10.22533/at.ed.1593352311057.
Escalante-Aburto, A., Mariscal-Moreno, R.M., Santiago-Ramos, D., Ponce-García, N., 2020. An update of different nixtamalization technologies, and its effects on chemical composition and nutritional value of corn tortillas. Food Rev. Int. 36, 456–498. https://doi.org/10.1080/87559129.2019.1649693.
Figueroa, J.D.C., Rodríguez-Chong, A., Veles-Medina, J.J., 2011. Proceso ecológico de Nixtamalización para la producción de harinas masas y tortillas integrales, 289. Mexican patent, pp. 1–22, 339.PA / a / 2005 / 011797.
Gallego-Castillo, S., Taleon, V., Talsma, E.F., Rosales-Nolasco, A., Palacios-Rojas, N., 2021. Effect of maize processing methods on the retention of minerals, phytic acid and amino acids when using high kernel-zinc maize. Curr. Res. Food Sci. 4, 279–286. https://doi.org/10.1016/j.crfs.2021.03.007.
Gasca-Mancera, J.C., Casas-Alencáster, N.B., 2007. Adición de harína de maíz nixtamalizado a masas fresca de maíz nixtamalizado. Ef. las Prop. Texturales Masa Tortilla 6, 317–328.
Hassan, S.M., Forsido, S.F., Tola, Y.B., Bikila, A.M., 2024. Physicochemical, nutritional, and sensory properties of tortillas prepared from nixtamalized quality protein maize enriched with soybean. Appl. Food Res. 4, 100383. https://doi.org/10.1016/j.afres.2023.100383.
Hernández-Salazar, M., Agama-Acevedo, E., Sáyago-Ayerdi, S.G., Tovar, J., Bello-Pérez, L.A., 2006. Chemical composition and starch digestibility of tortillas prepared with non-conventional commercial nixtamalized maize flours. Int. J. Food Sci. Nutr. 57, 143–150. https://doi.org/10.1080/09637480600659136.
Hernández-Uribe, J.P., Ramos-López, G., Yee-Madeira, H., Bello-Pérez, L.A., 2010. Physicochemical, rheological and structural characteristics of starch in maize tortillas. Plant Foods Hum. Nutr. 65, 152–157. https://doi.org/10.1007/s11130-010-0165-5.
Jane, J.-L., 1993. Mechanism of starch gelatinization in neutral salt solutions. Starch/Stärke 45 (5), 161–166. https://doi.org/10.1002/star.19930450502.
Mariscal Moreno, R.M., Figueroa, J.D.C., Santiago-Ramos, D., Villa, G., Sandoval, S.J., Rayas-Duarte, P., Véles-Medina, J.J., Martínez Flores, H.E., 2015. The effect of different nixtamalisation processes on some physicochemical properties, nutritional composition and glycemic index. J. Cereal Sci. 65, 140–146. https://doi.org/10.1016/j.jcs.2015.06.016.
Mariscal-Moreno, R.M., Chuck-Hernández, C., Figueroa-Cárdenas, J., de, D., Serna-Saldivar, S.O., 2021a. Physicochemical and nutritional evaluation of bread incorporated with ayocote bean (*Phaseolus coccineus*) and black bean (*Phaseolus vulgaris*). Processes 9, 1782. https://doi.org/10.3390/pr9101782.
Mariscal-Moreno, R.M., Figueroa-Cárdenas, J., de, D., Santiago-Ramos, D., Aguilar Arteaga, K., Flores Casamayor, V., Rincón-Aguirre, A., 2021b. Chemical, thermal and structural properties of alkaline-cooked (nixtamalised) chickpea (Cicer arietinum) flours. Int. J. Food Sci. Technol. 56, 3474–3483. https://doi.org/10.1111/ijfs.14974.
Mendez-Montealvo, G., Sánchez-Rivera, M.M., Paredes-López, O., Bello-Pérez, L.A., 2006. Thermal and rheological properties of nixtamalized maize starch. Int. J. Biol. Macromol. 40, 59–63. https://doi.org/10.1016/j.ijbiomac.2006.05.009.
Odukoya, J.O., De Saeger, S., De Boevre, M., Adegoke, G.O., Audenaert, K., Croubels, S., Antonissen, G., Odukoya, J.O., Njobeh, P.B., 2022. Influence of nixtamalization cooking ingredients on the minerals composition of nixtamalized maize and sorghum. J. Cereal Sci. 103, 103373. https://doi.org/10.1016/j.jcs.2021.103373.
Rincón-Aguirre, A., Figueroa-Cárdenas, J., de, D., Ramírez-Wong, B., Arámbula-Villa, G., Jiménez-Sandoval, S.J., Martinez-Flores, H.E., Pérez-Robles, J.F., 2021. Effect of nixtamalization with $Ca(OH)_2$, $CaCl_2$, and $CaCO_3$ on the protein secondary structure, rheological, and textural properties of soft wheat flour doughs. J. Cereal Sci. 101, 103271. https://doi.org/10.1016/j.jcs.2021.103271.
Sánchez-Villa, C.E., Zepeda-Bautista, R., Ramírez-Ortiz, M.E., Corzo-Ríos, L.J., 2020. Nixtamalized tortillas supplemented with proteins isolated from *Phaseolus coccineus* and huauzontle (*Chenopodium berlandieri* subsp. Nuttalliae) flour: rheological, textural, and sensorial properties. Int. J. Gastron. Food Sci. 22, 100274. https://doi.org/10.1016/J.IJGFS.2020.100274.
Santamaria, M., Garzon, R., Rosell, C.M., 2023. Impact of starch-hydrocolloid interaction on pasting properties and enzymatic hydrolysis. Food Hydrocoll. 142, 108764. https://doi.org/10.1016/j.foodhyd.2023.108764.
Santamaria, M., Garzon, R., Skendi, A., Papageorgiou, M., Rosell, C.M., 2025. Examining the role of polyphenols in the *in vitro* starch hydrolysis of gluten free enriched flatbreads. Food Chem. 486, 144456. https://doi.org/10.1016/j.foodchem.2025.144456.
Santiago-Ramos, D., Figueroa-Cárdenas, J., de, D., Véles-Medina, J.J., 2018. Viscoelastic behaviour of masa from corn flours obtained by nixtamalization with different calcium sources. Food Chem. 248, 21–28. https://doi.org/10.1016/j.foodchem.2017.12.041.
Statista, 2026. Mexico: corn tortillas production value 2014-2024. Feb 2026. https://www.statista.com/statistics/1044245/production-of-corn-tortillas-mexico.
Sunico, D.J.A., Rodriguez, F.M., Tuaño, A.P.P., Mopera, L.E., Atienza, L.M., Juanico, C. B., 2021. Physicochemical and nutritional properties of nixtamalized quality protein maize flour and its potential as substitute in Philippine salt bread. Chiang Mai Univ. J. Nat. Sci. 20. https://doi.org/10.12982/CMUJNS.2021.035.
Topete-Betancourt, A., Santiago-Ramos, D., Figueroa-Cárdenas, J., de, D., 2020. Relaxation tests and textural properties of nixtamalized corn masa and their relationships with tortilla texture. Food Biosci. 33, 100500. https://doi.org/10.1016/j.fbio.2019.100500.
Treviño-Mejía, D., Luna-Vital, D.A., Gaytán-Martínez, M., Mendoza, S., Loarca-Piña, G., 2016. Fortification of commercial nixtamalized maize (Zea mays L.) with common bean (*Phaseolus vulgaris L.*) increased the nutritional and nutraceutical content of tortillas without modifying sensory properties. J. Food Qual. 39, 569–579. https://doi.org/10.1111/jfq.12251.